

Assignment: NET Centric Computing

Course Code: CSC378

Topic: Unit 6 - State Management on ASP.NET Core Application (Theory & Practical)

Prepared by: Mr. Subash Pariyar

Due Date: 24 August

Submission Instructions:

Please complete the following questions by combining theoretical explanations with practical implementations. For practical components, include required commands, code snippets, configuration details, and screenshots of application execution and output results.

Questions:

1. Implementing Server-Side State Management in ASP.NET Core

Explain the concept of state management in ASP.NET Core applications and discuss why state management is required in stateless HTTP communication. Explain server-side state management strategies including Session State, TempData, HttpContext, and Cache with suitable examples.

Develop an ASP.NET Core MVC application to demonstrate server-side state management techniques.

Perform the following tasks:

1. Create an ASP.NET Core MVC application using .NET CLI.
2. Implement Session State to store and retrieve user information across multiple requests.
3. Implement TempData to transfer data between controller actions and explain how it differs from Session State and ViewData.
4. Use HttpContext to access request and response information.
5. Implement Caching to store frequently accessed data and demonstrate its benefits in improving application performance.
6. Configure required middleware for state management.
7. Demonstrate:
 - Storing data.
 - Retrieving data.
 - Removing stored data.
8. Document:

- Stateless HTTP and state management process.
- Server-side state management strategies.
- Difference between Session, TempData, HttpContext, and Cache.
- Screenshots of implementation and output.

2. Implementing Client-Side State Management in ASP.NET Core

Explain client-side state management techniques in ASP.NET Core applications. Discuss the purpose and usage of Cookies, Query Strings, and Hidden Fields with suitable examples.

Develop an ASP.NET Core MVC application to demonstrate client-side state management.

Perform the following tasks:

1. Create an ASP.NET Core MVC application.
2. Implement Cookies to:
 - Store user preferences or login information.
 - Retrieve stored cookie values.
 - Remove cookies when required.
3. Implement Query Strings to pass data between pages using URLs.
4. Create and use Hidden Fields in forms to:
 - Store temporary information.
 - Pass values between client and server without displaying them.
5. Demonstrate the difference between:
 - Cookies.
 - Query Strings.
 - Hidden Fields.
6. Test the application and observe client-side state handling.
7. Document:
 - Client-side state management workflow.
 - Usage scenarios of cookies, query strings, and hidden fields.
 - Security considerations.

- Screenshots showing implementation and output.